Simulating Pedestrian Accessibility for Mobility-Impaired Individuals on Urban Footpaths

An Agent-Based Modelling Approach for Indian Street Environments

Abstract

Pedestrian infrastructure in Indian cities is characterised by irregular footpath widths, frequent surface-level obstructions, and a significant deficit of kerb ramps — conditions that disproportionately affect wheelchair users. This study presents a conceptual agent-based model developed in NetLogo to simulate heterogeneous pedestrian behaviour on an obstructed urban footpath. Three agent types are defined — general pedestrians, wheelchair users, and elderly pedestrians. Sensitivity analysis with a fixed random seed across five ramp-count values (0, 5, 10, 15, 20) produced three findings: a sharp non-linear blockage threshold at five ramps (156 events at zero ramps, zero events at five); a saturation point beyond five ramps where additional provision produces no measurable improvement; and a persistent accessibility gap of 68–76 ticks that remains even at maximum ramp provision. The model provides a policy-relevant simulation framework aligned with India's Rights of Persons with Disabilities Act (2016).

Keywords: agent-based modelling, pedestrian accessibility, mobility impairment, urban footpaths, inclusive design, India, NetLogo, emergent behaviour, saturation point

1. Introduction

Accessible pedestrian infrastructure is a basic requirement for equitable urban participation. In Indian cities, footpaths along commercial streets are frequently occupied by hawker stalls, parked vehicles, and construction materials, making them difficult or impossible to navigate for wheelchair users. India's Rights of Persons with Disabilities Act (2016) requires accessible public infrastructure, but implementation has been uneven partly because planners lack practical simulation tools for evaluating how specific interventions affect mobility outcomes across different population groups.

Most accessibility assessments are static — they record whether features are present but cannot represent how those features interact with real pedestrian behaviour. Agent-based modelling addresses this by simulating individuals making decisions in a spatial environment and measuring what emerges at the system level. The key analytical property is emergence: system-level outcomes such as blockage thresholds are not written into agent rules but arise from repeated interactions across many time steps.

This project builds a simple ABM in NetLogo to ask one question: how does the number of kerb ramps affect the mobility outcomes of wheelchair users compared to other pedestrian types? The model is a proof-of-concept, not a calibrated prediction tool.

2. Research Objective

To simulate how pedestrians with different physical capacities navigate an obstructed urban footpath, and to measure how changes in ramp provision affect the mobility outcomes of wheelchair users relative to other pedestrian types, with obstacle layout controlled by a fixed random seed.

3. Methods

3.1 Modelling Approach

ABM was chosen because accessibility is not a property of a wheelchair user or a footpath in isolation — it emerges from the interaction between a specific person and a specific infrastructure configuration. NetLogo was chosen for its accessible syntax and built-in visualisation tools.

3.2 Environment

The simulation world is a 2D grid of patches representing footpath surface. Patch types: walkable surface, obstacle (encroachments), kerb (level change), and ramp (accessible kerb crossing). A fixed random seed (42) is set at the start of each run to ensure the obstacle layout is identical across all five configurations.

3.3 Agent Types

Agent type	Speed	Main behaviour	Kerb response
General pedestrian	1.2 m/s	Shortest path, avoids obstacles	No penalty
Wheelchair user	0.5 m/s	Seeks ramps, logs blockages if none found	Blocked without ramp
Elderly pedestrian	0.8 m/s	Avoids crowding, prefers clear paths	Partial slowdown

3.4 Behavioural Rules

- **If ahead patch is walkable** — move forward at agent speed
- **If ahead patch is obstacle** — rotate 45 degrees and retry up to 8 times
- **If ahead patch is kerb (wheelchair users only)** — scan for ramp within search radius; if found, detour at 60% speed; if not, record blockage event
- **If crowd exceeds threshold (elderly agents)** — reduce speed by 50%

4. Results

ramp-count	Total blockages	% blocked	Avg time — wheelchair	Avg time — general	Accessibility gap
0	**156**	**100%**	140.0 ticks	59.5 ticks	80.5 ticks
5	0	0%	132.5 ticks	59.5 ticks	73.0 ticks
10	0	0%	135.9 ticks	59.5 ticks	76.3 ticks
15	0	0%	128.0 ticks	59.5 ticks	68.5 ticks
20	0	0%	131.8 ticks	59.5 ticks	72.2 ticks

General pedestrian time is 59.5 ticks across all runs because random-seed 42 fixes the obstacle layout.

Zone	ramp-count	What happens
Exclusion	0	156 blockages, 100% blocked, accessibility gap 80.5 ticks
Transition	5	Zero blockages, gap stabilises at 73.0 ticks
Saturation	10–20	No further improvement, travel time within 128–136 ticks, gap 68–76 ticks

5. Discussion

5.1 Non-linear blockage threshold

The sharpest result is the transition between ramp-count = 0 and ramp-count = 5. At zero ramps, 156 blockage events were recorded, and every wheelchair user was blocked — 100%. At five ramps, blockages dropped to zero. This is a complete elimination in one step, not a gradual improvement. This threshold was not programmed into any agent rule. It emerged from the spatial relationship between agent movement paths and ramp positions along the kerb column.

5.2 Persistent accessibility gap

Even after blockages are eliminated at ramp-count = 5, wheelchair users take between 68 and 76 ticks longer than general pedestrians to complete the same corridor. The accessibility gap does not close with ramp provision alone. It reflects the underlying speed difference (wheelchair: 0.5 m/s vs general: 1.2 m/s) and the detour cost of reaching ramp positions.

5.3 Saturation point

From ramp-count = 5 onwards, wheelchair travel time stays between 128 and 136 ticks regardless of whether 5, 10, 15, or 20 ramps are present — a variation of less than 6%. Adding ramps beyond 5 produces no meaningful improvement. The system has reached a saturation point. The bottleneck is no longer ramp availability but agent walking speed and obstacle layout on the post-kerb section.

5.4 Policy implications

Three observations follow for urban infrastructure planning in Indian cities.

1. **Zero-ramp streets are fully exclusionary.** 156 blockage events at ramp-count = 0 means most wheelchair users cannot complete the corridor — directly contravening the RPwD Act (2016).
2. **Five ramps (15% kerb coverage) is the minimum viable target.** Concentrated provision above this threshold on key streets is more cost-effective than spreading too few ramps across many streets under AMRUT guidelines.
3. **Ramps are necessary but not sufficient.** The 68–76 tick gap requires complementary measures. The combination of ramp spacing, obstacle clearance, and footpath width that minimises this gap is a doctoral-level research question.

5.5 Limitations

- **Single run per configuration** — multiple runs with averaged outputs would produce more statistically robust estimates
- **Grid geometry** — does not capture real footpath irregularity, gradient, or surface texture
- **Literature-derived speeds** — not calibrated from Indian pedestrian populations
- **Agent homogeneity within types** — no intra-type variability in speed or assistive device
- **No social dynamics** — yielding, assistance, and social navigation absent

6. Conclusion

This model produces three findings that a static accessibility audit cannot: a non-linear blockage threshold at approximately 15% ramp coverage; a saturation points beyond which additional ramps produce no improvement; and a persistent mobility gap that ramps alone cannot close. These findings motivate a doctoral study at larger scale with real GIS data, calibrated agent parameters, and multiple simulation run per configuration.

References

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